

Deepfake Detection Implementation Handbook

HF-managed production revision

ARCHITECTURE

HF-MANAGED

EVIDENCE-FIRST

PRODUCTION GATES

Authoritative decision. VeriTrust executes no learned production inference in browser code, Vercel functions, platform workers, local runtimes, or VeriTrust-owned model servers. Learned production inference uses Hugging Face-managed infrastructure. The canonical hierarchy is HF-operated hf-inference when the exact model/task and all gates pass; then Hugging Face Inference Endpoint; then private Hugging Face repository plus Hugging Face Inference Endpoint. Third-party Hugging Face Inference Providers are DISABLED_BY_DEFAULT, outside the authoritative production path, and never an automatic fallback.

Status: implementation architecture and research package; no production performance claim. All UNKNOWN values remain explicit.

Prepared from the executable repository, seven historical PDFs, fresh official Hugging Face research, primary competitor sources and C2PA standards.

Contents and control

This is an Implementation Architecture Handbook, not a standalone executable specification. Critical invariants, model identities, deployment classes, evidence semantics, receipt fields, lifecycle states and companion references are generated from validated canonical structured sources; all provider availability is point-in-time and promotion-gated.

NORMATIVE IMPLEMENTATION COMPANIONS: `Architecture_Source_of_Truth.yaml`; `Model_Registry_Proposal.json`; `HuggingFace_Deployment_Matrix.csv`; `Implementation_Backlog.csv`; `Database_Target_Contract.md`; `Verification_Test_Plans.md`; `Migration_Roadmap.md`.

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Cross-document invariants

- HF-managed learned inference only; no normal third-party provider tier.
- Registry resolution precedes dispatch.
- Receipt identifies exact deployment, bundle and ordered learned components.
- `raw_output`, `raw_label`, `raw_score`, `normalized_label`, `normalized_score`, `calibrated_probability` and `calibration_version` remain distinct.
- Expected-evidence lifecycle uses the canonical 16-state enum: `planned`, `dispatched`, `running`, `completed`, `abstained`, `not_applicable`, `unavailable`, `timed_out`, `failed`, `cancelled`, `privacy_blocked`, `policy_skipped`, `insufficient`, `unsupported`, `degraded`, `budget_exceeded`.
- Specialist evidence cannot silently substitute or own final policy.
- No original PDF or production source file is overwritten.

1. Executive decision and current code path

A full frame is not a face crop. Crop failure must produce failed/not-applicable face evidence, never a silent task change.

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Current `/api/v1/deepfake` calls a public HF Space for cropping and then a mutable Pixel/Prism alias. Crop/no-face/timeout returns the original frame; Pixel and Prism may substitute. The normalizer guesses label meaning.

2. Scientific role decomposition

Face manipulation, full-frame synthesis, independent low-level forensics, quality and OOD answer different questions. They remain separate evidence even when Gateway policy considers them together.

Role	Question	Target
Face manipulation	Is a qualified face crop manipulated?	private Yermandy HF Endpoint
Full-frame synthesis	Is the whole image likely generator-produced?	Community Forensics 384 HF Endpoint
Same-family shadow	Does a lower-resolution family alternative agree?	Community Forensics 224 HF shadow Endpoint
Independent forensics	Do residual/SRM/DCT-style low-level signals add error diversity?	untrained private model on HF Endpoint
OOD	Is inference within qualified support?	private SigLIP2-based head on HF
Quality	Are bytes/resolution/compression sufficient?	deterministic VeriTrust

3. HF-managed face detection, landmarks and alignment

A reviewed private HF Endpoint receives the full image, runs pinned OpenCV Zoo YuNet for boxes and five landmarks, applies versioned five-point similarity alignment, produces bounded stable-ID crops, and runs pinned Yermandy per crop. The canonical deployment bundle is `deepfake.face_yunet_yermandy.bundle.v1`; `inference_receipt.v2` preserves both learned components and their execution order.

- Zero faces means face manipulation not_applicable.
- Detection/crop failure means failed or insufficient.
- The full frame is never substituted as a face crop.

Order	Component role / model key	Repository / revision	Preprocessing	Qualification
1	face_detection_landmarks_alignment deepfake.face_pipeline.yunet_private_v1	veritrust-private/deepfake-face-pipeline-yunet-v1 @ UNKNOWN	face_alignment.v1	requires_packaging_and_qualification
2	face_manipulation deepfake.face_manipulation.yermandy_v1	yermandy/deepfake-detection @ 9a6857ec642deb57373c5437be803a199468b8c6	yermandy_face_crop_contract.v1	research_candidate

4. Target HF endpoint topology

The yermandy custom CLIP/PEFT path has no provider mapping and requires a controlled private copy/handler. Community Forensics uses a custom mixin/preprocessor and targets HF Endpoints. No component loads model weights in a VeriTrust worker.

VERITRUST ORCHESTRATOR
|- HF FACE ENDPOINT (private/custom)
|- HF FULL-FRAME ENDPOINT (protected/custom)
|- HF FORENSIC SHADOW ENDPOINT
|- HF PRIVATE OOD ENDPOINT

5. Media ingestion and quality

Accept JPEG/PNG/WebP through a signed upload or bounded request. Verify MIME, magic bytes, dimensions, pixel count, decompression bounds and SHA-256 before HF dispatch. Orientation, resize/crop and normalization are versioned. Quality observations are evidence, not fake/real labels.

- Reject active/polyglot/unsupported formats.
- Strip operational metadata only according to declared preprocessing while preserving the immutable original.
- Never log media or signed URLs.

6. Zero, one and multiple faces

Observation	Face role	Other roles
Zero faces	not_applicable	full-frame and quality may run
One face	bounded qualified crop	full-frame and OOD run independently
Multiple faces	stable face IDs; bounded per-face crops	policy must not hide minority-face risk
Crop failure	failed/insufficient	full-frame is separately identified

7. Input, output, registry and receipt

The face input schema requires crop bytes, parent artifact hash, face ID/bounds, preprocessing version and quality. Outputs must have an explicit declared label map and finite raw scores. Every response is bound to a registry resolution, deployment bundle and ordered component receipt.

Evidence	Required separation
Required fields	evidence_id, investigation_id, artifact_id, specialist, scientific_role, registry_resolution_id, deployment_id, model_key, immutable_model_revision, preprocessing_version, input_hash, raw_output, raw_label, raw_score, normalized_label, normalized_score, calibrated_probability, calibration_version, quality, sufficiency, ood_status, status, reason_codes, started_at, completed_at
Lifecycle states (16)	planned, dispatched, running, completed, abstained, not_applicable, unavailable, timed_out, failed, cancelled, privacy_blocked, policy_skipped, insufficient, unsupported, degraded, budget_exceeded
Semantic rule	raw_score, normalized_score, and calibrated_probability are separate nullable fields and never silently substituted.

Receipt	Required separation
inference_receipt.v2 fields	receipt_id, registry_resolution_id, deployment_id, deployment_bundle_id, request_id, provider, endpoint_identity, endpoint_deployment_revision, request_sha256, response_sha256, provider_request_id, components, attempt, latency_ms, billed_units, completion_status, received_at
components[] fields	component_role, model_key, repository_id, immutable_revision, artifact_or_container_digest, preprocessing_version, input_schema_version, output_schema_version, execution_order, qualification_status
Compatibility	A valid v1 receipt is exposed as a v2 receipt with one synthetic component derived from its singular repository/model/revision/preprocessing fields; missing historical provenance remains explicit and is never fabricated.

8. Normalization, calibration and disagreement

No threshold in this handbook is a production threshold; numeric gates require signed VeriTrust benchmark artifacts.

Model logits/similarities are not probabilities. Calibrate per role/revision/preprocessing on a held-out set; report Brier, ECE, reliability and slice confidence intervals. Strong disagreement or OOD produces abstention/review, not forced consensus.

9. Benchmark and robustness

- Cross-dataset: FaceForensics++, Celeb-DF, DFDC, FFIW and legally approved current corpora.
- Generator-family, manipulation-method, demographic, quality, compression, resize, crop and screenshot slices.
- Near-duplicate/group/identity leakage controls and temporal holdout.
- Adversarial transformations, metadata stripping and provider failure.
- Report false-positive/negative cost, abstention, OOD and calibration-not accuracy alone.

10. Privacy, failure and endpoint configuration

Use protected/private HF endpoints, scoped tokens and minimized role-specific bytes. Confirm region, retention/log terms and DPA. Keep min replicas only for latency-critical roles; asynchronous shadows

may scale to zero if cold-start behavior is handled.

Failure	Behavior
HF fully unavailable	High-impact: fail closed or mandatory review. Advisory: degraded/uncertain. Queue only when policy and deadline permit.
One endpoint unavailable	Retry only the same immutable deployment within budget; never silently change model/provider.
Malformed/unknown output	Reject schema/label; persist failed evidence; never guess from text or label digits.
Missing calibration	May display uncalibrated evidence; cannot authorize allow/block.
Timeout/cancel	Persist terminal status and preserve partial evidence; no invented benign value.

11. Cost, scaling and observability

Meter image count, bytes, endpoint minutes, retries and per-face fan-out. Bound faces per image and concurrency per tenant. Observe queue age, endpoint cold starts, preprocessing rejection, role sufficiency, disagreement, OOD, slice drift and cost.

- No retry to a different model.
- Deduplicate by artifact hash + role + exact contract where policy permits.
- Alert when crop/no-face rates change unexpectedly.

12. Tests, deployment and rollout

Unit tests cover bytes, quality, geometry, labels and schema. Contract tests use immutable golden requests at each HF endpoint. Integration tests cover signed upload/poll/webhook. Load tests include multi-face fan-out. Security tests include malicious images, response caps and secret/egress controls.

- Shadow with no decision authority.
- Canary only after benchmark/calibration/privacy/security/license gates.
- Rollback registry pointer atomically; retain original receipts.

13. Implementation backlog and done

- VT2-004 remove crop fallback; VT2-020 yermandy endpoint; VT2-021/022 Community Forensics; VT2-023 OOD; VT2-030/031 benchmark/calibration; VT2-038 production promotion.
- Done means each role has exact semantics, HF deployment, receipt, qualified calibration, explicit failure and no local learned runtime.

References and package evidence

Primary sources and immutable model snapshots are listed in Research/Sources/source_ledger.csv and Research/Sources/HuggingFace/. Repository claims are traceable through repository_inventory.csv, source_file_audit.csv, runtime_call_graph.md, current_architecture_map.md and code_claim_ledger.csv.

- Architecture_Source_of_Truth.yaml - canonical invariants, roles, schemas and rollout.
- Model_Registry_Proposal.json - immutable role/deployment proposal.
- HuggingFace_Deployment_Matrix.csv - point-in-time provider and endpoint decisions.
- Technical_Findings_Register.csv / Implementation_Backlog.csv - evidence-backed remediation.
- Database_Target_Contract.md / Database_Migration_Specification.md - logical contract without fabricated SQL.
- Competitor_Analysis.md / Competitor_Matrix.csv - primary-source market design loop.

Authoritative external references

Hugging Face Inference Providers | Hugging Face Inference Endpoints | C2PA Technical Specification. Access dates, exact page URLs, claim summaries and verification status are preserved in the source ledger.

Universal operational controls

Security, privacy, cost, capacity, calibration, observability, incident response and rollback are release gates for every role. No UNKNOWN is replaced with an estimate; unresolved values remain explicit until an evidence-producing test or owner decision closes them.

Final boundary

Hugging Face-managed learned inference is authoritative. No learned production model runs inside the VeriTrust Vercel application or a VeriTrust-owned worker/server; no production weights are required in the VeriTrust deployment repository.